IF THIS PAGE IS STAPLED, USE SCISSORS TO CUT THE STAPLED CORNER

Number and hand in all subsequent pages along with any additional solution sheets

Data structures and algorithms, exam

(DAT495, DAT038, DAT525, DIT183, DIT962, TDA417)

Wednesday 2026-10-28, 14:00–18:00

Teachers: **Peter Ljunglöf, 031-772 1065 (primary phone contact during the exam)**
Jonas Duregård, 031–772 1028
Yehia Abd Alrahman
Christian Sattler

We will visit each exam room at least twice.

Answer directly on the question sheets!
If you need additional space (e.g., for the advanced questions), you can use extra pages.
Refer to them from the question sheet so that we don't miss them, and number both the printed and extra pages in a correct sequence. Include the printed pages in your page count.

Write your anonymous code on each printed question sheet, and on all extra pages.

You may answer in English or Swedish.
You may write algorithms in your style of pseudocode or your favourite programming language.

Excessively complicated answers may be rejected.
Write legibly, we need to be able to read your answer!

Questions and solutions will be published afterwards here:
https://github.com/ChalmersGU-data-structure-courses/past-exams

Allowed aids One *hand-written* A4 sheet of notes (double-sided).
If you bring such a sheet, take it with you after the exam.

Review When the exams have been graded, they will be available for review in Canvas.
If you want to discuss the grading, contact your examiner via email with a copy of your exam.
Remarks about the grading should be explained thoroughly.

Grading There are 6 basic questions, each graded **correct** or **incorrect** and worth 1 basic point.
In corner cases, we may also give half points. To pass, you must obtain 5 of the 6 basic points.

There are 3 advanced questions, each worth 2 advanced points. For a higher grade, you must pass and also obtain a certain number of the 6 advanced points:

- **grade 4**: at least 2 advanced points
- **grade 5**: at least 4 advanced points

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	Anonym kod		Question no. Uppgift nr **1**

Basic question 1: Searching and sorting

Here is a sorted array with 15 integer values:

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
10	30	40	50	70	80	90	100	120	140	150	170	180	200	210

a) A binary search is performed, searching for the value 71. Indices low and high are used to specify the search interval. What are the values of low and high after each comparison?

Note: There are more cells than you need here! You should fill in exactly one column per comparison performed by the search. The initial values of high and low are pre-filled.

Number of compared elements:	0	1	2	3	4	5	6	7	8
Low:	0								
High:	14								

b) Are there any values in the array that can be found using fewer comparisons with a linear search from left to right, than with a binary search? List **all** such values.

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Data structures and algorithms, written exam	2026-10-28